

OSPF ADVANCED

1. PROJECT OVERVIEW

This project showcases the design and deployment of a highly available, scalable, and secure multi-site Enterprise Network infrastructure. It integrates advanced dynamic routing behaviors, precise multi-protocol redistribution policies, secure area adjacencies, and intelligent first-hop gateway redundancy.

2. ROUTING & ADVANCED TRAFFIC ENGINEERING

The core enterprise backbone runs multiple optimized interior routing domains with controlled traffic interaction:

- **Multi-Area OSPF:** Serves as the primary enterprise routing core, segmented into Area 0 (Backbone), Area 1, and Area 2 to maintain structured hierarchy and reduce CPU computational overhead.
- **OSPF Authentication:** Configured on OSPF routing links to secure neighbor adjacencies, preventing unauthorized routing updates and mitigating potential malicious injection attacks.
- **OSPF Virtual-Link:** Since Area 2 is geographically disconnected from Area 0, an engineered OSPF Virtual-Link is established through Area 1 to extend transit capabilities and ensure proper database synchronization.
- **EIGRP (AS 100) & RIP Domains:** Deployed at respective branch environments to ensure rapid internal convergence (EIGRP) and compatibility with legacy remote nodes (RIP).
- **Redistribution & Route-Map Filtering:** Executed at Autonomous System Boundary Routers (ASBRs) with precise `route-map` policies configured to strictly deny specific subnets (e.g., `net 5.5.5.5`) from being injected across routing boundaries, preventing suboptimal paths and loop behaviors.

3. HIGH AVAILABILITY & GATEWAY FAILURE INFRASTRUCTURE

To eliminate single-point-of-failure vulnerabilities at the access layers, intelligent network health tracking is implemented:

- **VRRP (Virtual Router Redundancy Protocol):** Configured at the distribution layer of remote sites to provide a stable Virtual IP gateway (e.g., VIP: 192.168.1.100) across host segments.
- **IP SLA & Object Tracking:** The primary Master VRRP node continuously gauges upstream WAN link health and external destination reachability utilizing persistent IP SLA probes.
- **Dynamic Decrement Priority Failover:** Upon detecting upstream path latency or physical link degradation, the tracked object programmatically triggers a ****Priority Decrement**** on the primary gateway. This forces an immediate, transparent transition of the Backup router to ****Master**** state, preserving end-to-end traffic flows.

4. NETWORK INFRASTRUCTURE SERVICES

- **Dynamic Host Addressing:** A dedicated, central DHCP Router is provisioned within the EIGRP domain to automate IP, mask, and option allocations for endpoint devices.
- **Host Integration:** The infrastructure is bound to a physical host network utilizing a real-world VMware Network Adapter (VMnet8) to simulate live virtualization or external server environment handoffs.
- **Stable Management Offsets:** Loopback interfaces (e.g., 150.0.0.x) are declared on every active router node to act as immutable Router IDs and assure persistent remote management accessibility.

Technical Documentation - Completed